

Chapter 1: Overview of Obstetrics — Revision Notes

Williams Obstetrics 26e, pp. 28–51 of the PDF. High-yield numbers are in **bold**. The two tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included from the PDF.

Big picture: Obstetrics is judged by maternal and perinatal outcomes. Know the **standard vital-statistics definitions** cold, the **US numbers** (perinatal mortality, maternal mortality ratio, SMM), the **leading causes of maternal death**, and the current "hot topics" (COVID-19, maternal mortality programs, opioids, prenatal genetics, placenta accreta spectrum, 17-OHPC).

1. Vital Statistics

- The **National Vital Statistics System** collects US births and deaths, including fetal deaths. Legal authority rests with the **50 states**, 2 regions (DC, New York City) and 5 territories.
- The standard birth certificate records medical/lifestyle risks, labor and delivery factors and newborn characteristics.
- The current **death certificate has a pregnancy checkbox**.
- Most evidence in obstetrics is **not high quality** → much practice rests on expert opinion and bodies such as ACOG, SMFM, CDC and NIH.

Definitions

- Standard definitions come from **WHO, AAP and ACOG**; uniformity allows comparison between regions and countries.
- ⚠ **Fetal-death reporting is not uniform** across states:
 - 28 states:** losses from **20 weeks** are fetal deaths.
 - 8 states:** all products of conception.
 - Others: a minimum birthweight of **350, 400 or 500 g**.
- National Vital Statistics Reports tabulate fetal deaths at **≥20 weeks**.
- The 50th-centile fetal weight at **20 wk ≈ 325–350 g**, much less than 500 g. **500 g ≈ 50th centile at 22 wk**.

Births and perinatal period

Term	Definition
Perinatal period	Birth of a neonate after 20 wk → 28 completed days after birth. If based on birthweight, it starts at the birth of a 500-g neonate.
Birth	Complete expulsion or extraction of a fetus after 20 wk . Without accurate dating, fetuses <500 g are abortuses , not births.
Birthweight	Weight immediately after delivery, to the nearest gram .
Live birth	Newborn breathes spontaneously or shows any other sign of life (heartbeat, definite spontaneous voluntary movement). Transient cardiac contractions and fleeting gasps don't count.
Stillbirth / fetal death	Absence of signs of life at birth.
Abortus	Fetus/embryo removed or expelled in the first half of gestation: ≤20 wk , or <500 g if dating is uncertain.
Induced termination	Purposeful interruption of an intrauterine pregnancy intended not to produce a liveborn neonate, and not resulting in a live birth. Excludes retained products after fetal death.

Rates

Rate	Numerator	Denominator
Birth rate	Live births	1000 population
Fertility rate	Live births	1000 females aged 15–44
Stillbirth (fetal death) rate	Stillbirths	1000 births (live + stillbirths)
Neonatal mortality rate	Neonatal deaths	1000 live births
Perinatal mortality rate	Stillbirths + neonatal deaths	1000 total births
Infant mortality rate	Infant deaths (birth to 12 months)	1000 live births
Maternal mortality ratio	Maternal deaths from the reproductive process	100,000 live births

- **Early neonatal death:** first **7 days**. **Late neonatal death:** after 7 days but **before 29 days**.
- **Why "ratio", not "rate":** the numerator counts deaths regardless of pregnancy outcome (live births, stillbirths, ectopics), but the denominator is only **live births**.

Birthweight

Category	Weight
Low birthweight (LBW)	<2500 g
Very low birthweight (VLBW)	<1500 g
Extremely low birthweight (ELBW)	<1000 g

Gestational age

Category	Gestational age
Preterm	Before 37 completed weeks (the 259th day)
— Early preterm	Before 34 completed weeks
— Late preterm	34–36 completed weeks
Term	37 to 42 completed weeks (260–294 days)
— Early term	37⁰/₇ – 38⁶/₇ wk
— Full term	39⁰/₇ – 40⁶/₇ wk
— Late term	41⁰/₇ – 41⁶/₇ wk
Postterm	After completion of the 42nd week (from day 295)

Maternal deaths

Term	Definition	Example
Direct	From obstetrical complications of pregnancy, labor or puerperium, or from interventions, omissions, incorrect treatment or a chain of events from these	Exsanguination after uterine rupture
Indirect	Not directly obstetrical: pre-existing or new disease aggravated by the physiology of pregnancy	Mitral stenosis
Late maternal	Direct or indirect cause, >42 days but <1 year after pregnancy ends	—
Nonmaternal	Accidental or incidental, unrelated to pregnancy	Car accident, concurrent malignancy
Pregnancy-associated	Death from any cause while pregnant or within 1 calendar year of pregnancy end, regardless of duration or site	—
Pregnancy-related	A pregnancy-associated death from (1) complications of pregnancy, (2) a chain of events started by pregnancy, or (3) aggravation of an unrelated condition by pregnancy's physiological or pharmacological effects	—

Memory hook: "Associated = **A**ny cause; Related = **R**esulting from pregnancy." Every pregnancy-related death is pregnancy-associated, not vice versa.

2. Pregnancy Rates in the United States

- **Fertility rate (2019): 58 live births per 1000 women** aged 15–44. Falling since 1990 and now **below replacement** → population decline.
- Birth rate ↓ in all major racial/ethnic groups, adolescents, unmarried women and ages 20–24. **Slightly ↑ in women >30.**
- Almost **half of 2019 newborns were minorities:** Hispanic **25%**, African-American **15%**, Asian **4%**.
- **45% of US births are unintended** at conception (Guttmacher), though the proportion has fallen since 2008.
 - More likely in unmarried and black women, and in women with less education or income.

Table 1-1. Total Pregnancies and Outcomes in the United States in 2019

Outcome	Number or Percent
Total births	3,747,540
— Cesarean deliveries	31.7%
— Primary cesarean delivery	21.6%
— Vaginal birth after cesarean	13.8%
— Preterm births (<37 weeks)	10.0%
— Low birthweight (<2500 g)	8.0%
— Very low birthweight (<1500 g)	1.4%
Induced abortions	862,320

Data from Guttmacher 2019b; Martin, 2021.

3. Obstetrical Care Measures

Perinatal and infant mortality

- **Perinatal mortality rate (2016): 6 per 1000 births**, unchanged for several years.
- Fetal deaths at **≥28 wk have declined** since 1990; those at **20–27 wk are static.**
- **Infant mortality: ~6 per 1000 live births (2018)**, vs nearly 7 in 2001.
- **Four leading causes of infant death** (together almost **half**):
 1. **Congenital malformations**
 2. **Preterm birth**
 3. **Low birthweight**
 4. **Maternal pregnancy complications**
- **17%** of infant deaths in 2018 were in neonates born **preterm and LBW.**
- Intensive care can now be offered to some neonates **<500 g** (Ch 45).

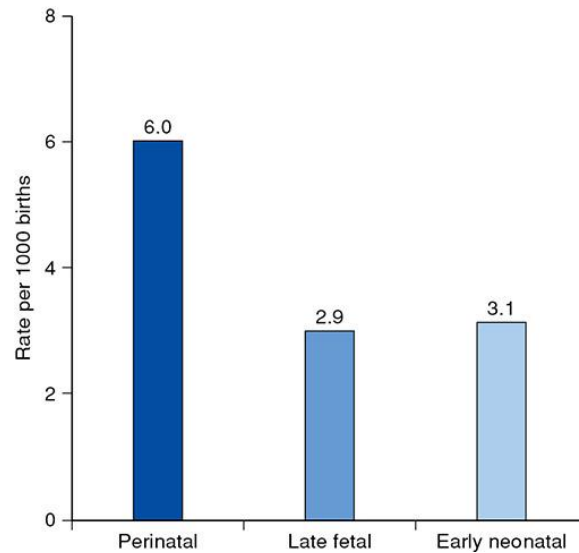


Figure 1-1. US 2016: perinatal mortality 6.0/1000 births = late fetal 2.9 + early neonatal 3.1.

Maternal mortality

- Fell precipitously in the 20th century; now measured **per 100,000 births**.
- The CDC's **Pregnancy Mortality Surveillance System (PMSS)**: **3410** pregnancy-related deaths in 2011–2015.
 - **~5%** were early-pregnancy deaths (ectopic or abortive outcomes).
- **Causes:**
 - **Obstetrical triad (hemorrhage, preeclampsia, infection)** → **⅓** of all deaths.
 - **Thromboembolism, cardiomyopathy and other cardiovascular disease** → **another ⅓**.
 - **Cerebrovascular accidents 8.2%, amnionic fluid embolism 5.5%.**
 - **Anesthesia-related only 0.4%** (all-time low).

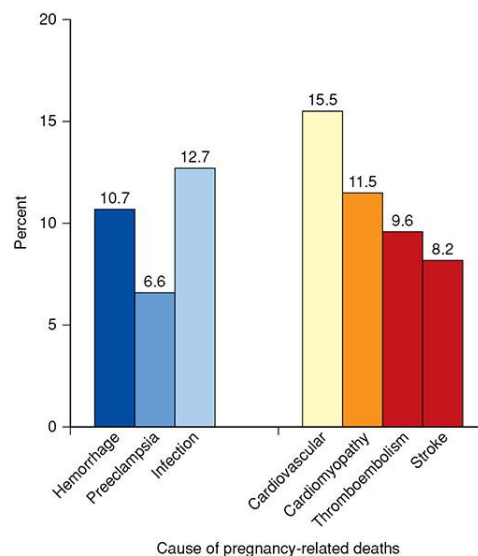


Figure 1-2. Causes of US pregnancy-related deaths 2014–2017: cardiovascular 15.5%, infection 12.7%, cardiomyopathy 11.5%, hemorrhage 10.7%, thromboembolism 9.6%, stroke 8.2%, preeclampsia 6.6%.

- **Pregnancy-related maternal mortality ratio: 17 per 100,000 live births (2017)**, rising over the last 30 years.
- **Why the apparent rise:**
 1. More women with **severe chronic health conditions**
 2. More births to women **>40 years**
 3. Artificial elevation from **ICD-10** (implemented 1999)
 4. **Better reporting**
 5. **Pregnancy checkbox** on the death certificate

- After accounting for the checkbox, predicted maternal mortality **did not change significantly from 1999 to 2017**.

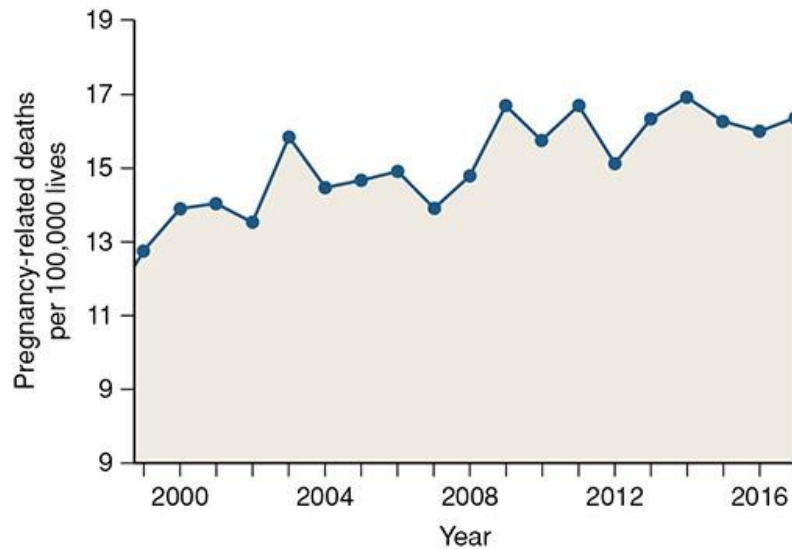


Figure 1-3. US pregnancy-related mortality 1999–2017: an upward trend from ~12–13 to ~17 per 100,000.

- **Racial disparity:** black women have the highest ratio; causes are health-care availability, access and use. Rates are also higher in **rural** than metropolitan areas.

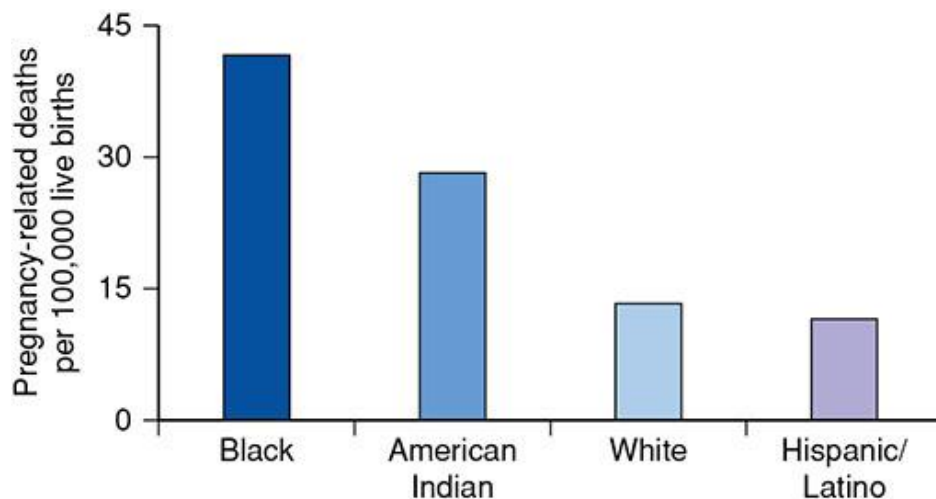


Figure 1-4. Pregnancy-related mortality by race/ethnicity 2014–2017: Black ~41, American Indian ~28, White ~13, Hispanic ~11 per 100,000.

- **Preventability:**
 - Up to **1/3 of deaths in white women** and up to **1/2 in black women** (Berg, 2005).
 - **28%** of 98 deaths in an insured cohort (Clark, 2008).

Severe maternal morbidity (SMM)

- **Definition:** unintended events of labor and delivery with **serious short- or long-term consequences** to the woman.
- ACOG and SMFM (2016) give suggested screening topics.
- CDC analysis of >50 million records (1998–2009): **129 per 10,000** women had ≥ 1 SMM indicator.
- **~200 women have SMM for every maternal death.**
- SMM rates have **risen over 15 years**, attributed to better documentation and a **rise in blood transfusion**.
- Highest in **smaller hospitals (<1000 deliveries/yr)**. Black women are disproportionately affected.

Table 1-2. Severe Maternal Morbidity Indicators

Acute myocardial infarction	Disseminated intravascular coagulation	Pulmonary edema
Acute renal failure	Eclampsia	Severe anesthesia complications
Adult respiratory distress syndrome	Heart failure during procedure	Sepsis
Amnionic fluid embolism	Hysterectomy	Shock
Cardiac arrest/ventricular fibrillation	Injuries of thorax, abdomen, and pelvis	Sickle-cell crisis
Cardiac monitoring	Intracranial injuries	Thrombotic embolism
Cardiac surgery	Puerperal cerebrovascular disorders	Tracheostomy
Conversion of cardiac rhythm		Ventilation

Summarized from the Centers for Disease Control and Prevention, 2021. (The printed table is a single alphabetical list of 23 indicators, shown here in three columns.)

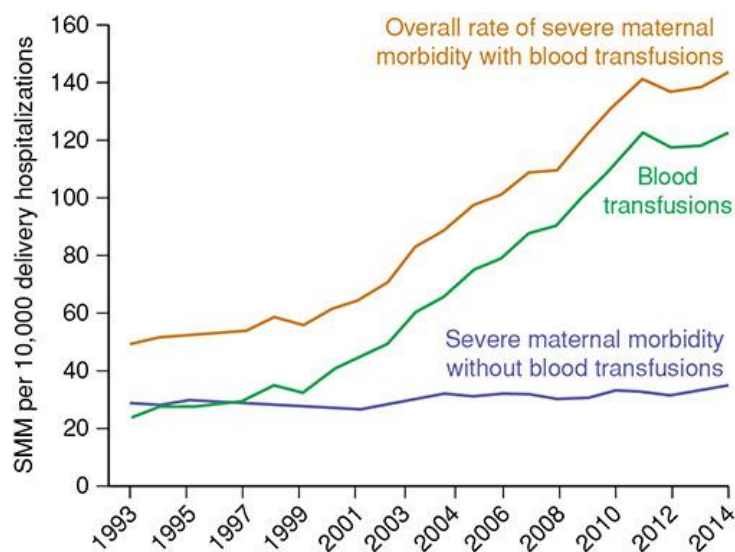


Figure 1-5. SMM per 10,000 delivery hospitalizations 1993–2014: the rise is driven by blood transfusion; SMM without transfusion is flat (~30).

Near misses

- **Near miss / close call:** an **unplanned event caused by error** that **does not injure** the patient but could have.
- More common than injury events, but harder to identify and quantify. Reporting systems allow focused safety work.
- Systems:
 - **WHO** near-miss system (validated in Brazil; correlates with maternal death rates)
 - **UKOSS** (UK Obstetric Surveillance System)
 - **National Partnership for Maternal Safety** (USA)

4. Timely Topics in Obstetrics

COVID-19 pandemic

- SARS-CoV-2, early 2020: the greatest public health crisis since the **1918 influenza pandemic**.
 - By early 2021: >181 million infected, nearly 4 million deaths worldwide.
 - US 2020: >500,000 deaths, including >3000 healthcare workers.
- Prenatal care changed: **virtual and drive-through** models.

- Asymptomatic or mild infection is common in pregnancy. The effect of pregnancy on the disease course is **controversial**.
- **mRNA vaccines are safe and highly effective**, but the data came late because pregnant women were **excluded from the initial trials**. The FDA approved COVID-19 vaccines for pregnant women in **2021**.
- **Chinn (2021): 2.2%** of >800,000 pregnancies had COVID-19 → ↑ **preterm birth, ICU admission, intubation/ventilation and maternal death**.
- A combined in-person plus **audio-only** virtual care model may best reach vulnerable women without internet access.

Maternal mortality — a call to arms

- **~700 US women die each year** from pregnancy or its complications; many deaths are preventable.

Program	Role
PMSS (CDC)	National pregnancy-related death data to guide prevention
AIM (Alliance for Innovation on Maternal Health)	Patient safety bundles : evidence-based best practices
The Joint Commission	Birthing centers should have protocols and simulation
National working groups	Target specific risks, e.g., thromboembolism

- The **puerperium** is also vulnerable → dedicated **1-year postpartum follow-up** (hypertension, diabetes, cardiovascular disease). The **"fourth trimester"** concept (Ch 36).
- **MOMMA's Act**: extend Medicaid postpartum coverage from **60 days to 12 months**.

Opioid use disorder

- Prevalence in pregnancy ↑ **333%** from **1.5 to 6.5 per 1000 deliveries** (1999–2014).
- → Unprecedented rise in **neonatal opioid withdrawal syndrome** (Ch 33).
- **ACOG: universal screening by questionnaire + multidisciplinary care**; curtail therapeutic opioid use.
- Treatment: **methadone or buprenorphine** (Ch 64).

Advances in prenatal genetics

Test	Key points
cfDNA screening	Noninvasive; now commonplace
Chromosomal microarray (CMA) on CVS or amniotic fluid	Detects gains/losses as small as 50–100 kb . Higher yield than karyotype, but most birth defects have a normal CMA and karyotype .
Next-generation sequencing panels	Targeted genes, e.g., for skeletal dysplasia
Whole exome sequencing (WES)	All coding regions = 1.5% of the genome . Finds a significant abnormality in ~10% of structurally abnormal fetuses with normal CMA and karyotype, and ~30% of unexplained nonimmune hydrops .

- **WES is not recommended for routine prenatal diagnosis** (ACOG). Limitations: many **variants of uncertain significance**, long turnaround, high cost, and **incidental medically actionable findings** → comprehensive counseling.

Placenta accreta spectrum (PAS)

- Cesarean rate is **static at ~32%**, but **PAS has grown**: up to **1 in 300 deliveries**.
- Responses: specialized **accreta surgical teams** at tertiary centers, more antepartum transfer, and efforts to **avoid the primary cesarean**.
- PAS will likely remain a significant cause of SMM (Ch 43).

Progestogens to prevent preterm birth

- **17-OHPC** (17-alpha-hydroxyprogesterone caproate, IM; **Makena**): FDA **accelerated approval**, contingent on a second confirmatory trial.
- **PROLONG trial (2019): no benefit vs placebo** for birth <35 wk.
- Timeline:
 - 2019: FDA Advisory Committee voted **9 to 7** to withdraw approval.
 - Late 2020: **FDA CDER recommended withdrawal**, and still does (2021).
- **ACOG and SMFM continued to endorse 17-OHPC**, provided the **"uncertainty regarding benefit"** is shared with the patient.
- **EPPIIC meta-analysis (2021)**: progestogens (including 17-OHPC) reduced births <34 wk, but **not statistically significantly**.

Quick-Recall: Denominators and Cutoffs

Measure	Per	Cutoff / window
Birth rate	1000 population	—
Fertility rate	1000 women 15–44	US 2019: 58
Stillbirth rate	1000 total births	≥20 wk (or ≥500 g)
Perinatal mortality rate	1000 total births	>20 wk birth → 28 days; US: 6
Neonatal mortality	1000 live births	Early ≤7 d; late >7 d to <29 d
Infant mortality	1000 live births	Birth → 12 months; US: ~6
Maternal mortality ratio	100,000 live births	US 2017: 17
Pregnancy-associated death	—	Any cause, pregnancy → 1 year
Late maternal death	—	>42 days to <1 year
Abortus	—	≤20 wk or <500 g
LBW / VLBW / ELBW	—	<2500 / <1500 / <1000 g
Preterm / term / postterm	—	<37 / 37–42 / >42 wk

Key Numbers to Memorize

- Birth = after **20 wk**; abortus **<500 g**; 500 g ≈ 50th centile at **22 wk**
- Term **260–294 days**; preterm before day **259**; postterm from day **295**
- Fertility rate **58/1000** (below replacement); **45%** of births unintended
- 2019: **3.75 million** births; cesarean **31.7%**; preterm **10%**; LBW **8%**; VLBW **1.4%**
- Perinatal mortality **6/1000**; infant mortality **~6/1000**
- Maternal mortality ratio **17/100,000**; **~700** US deaths/yr; black women **~3x** white
- Causes: hemorrhage + preeclampsia + infection **⅓**; thromboembolism + cardiomyopathy + CVD **⅓**; stroke **8.2%**; AFE **5.5%**; anesthesia **0.4%**
- SMM **129/10,000**; **~200 SMM cases per maternal death**
- Opioid use disorder **1.5 → 6.5/1000** deliveries (**+333%**)
- CMA resolution **50–100 kb**; WES yield **~10%** (hydrops **~30%**); exome = **1.5%** of genome
- PAS up to **1 in 300**; cesarean rate **~32%**